

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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We study minimum-cost revision of an installed policy with operating protection after every state–date restart and one common randomized Markov continuation. We characterize the supremum of restart-price certificates by an occupation-flow linear program and identify the shared successor-budget equations that restore the original constrained problem. A strict separation example distinguishes the price relaxation from both a constant-multiplier bound and the constrained optimum. An end-to-end finite algorithm constructs operating enclosures, feasible policies, lower certificates, and covering refinements without access to exact optimal operating values or inherited incumbents. Its accuracy rule adapts arithmetic precision. For controlled finite-atomic transitions and a finitely supported initial law, an exact restriction–extension theorem transfers paired finite certificates to the full Borel-state problem. A source-frozen evaluation produces four target hits in twelve new primary environments for each of two independently checked formulations; the price formulation tightens all eight unclosed intervals. Two positive-cost observed-fleet instances of the original affine dynamics have paired continuum certificates. These finite-initial-law results do not close the thirty earlier positive-cost intervals under the uniform initial law. All runs, costs, and remaining intervals are reported.

KEYWORDS: Dynamic programming, Policy revision, Common Markov policies, Constrained control, Verified computation.

1. INTRODUCTION

An organization may promise operating protection whenever a decision rule is restarted, while evaluating the expense of implementing that rule under its actual initial population. These are different restrictions on a single policy. The first is a statewise performance contract; the second is an ex ante cost objective. Actions alter future states and therefore alter future implementation expense. A collection of inexpensive plans, each optimized for a different restart, need not define one policy at their common successor states.

We retain this distinction throughout. The decision variable is a Borel randomized Markov policy with one action distribution at each state and date. Its randomization is fresh conditional on that state and date; a hidden initial lottery over persistent plans is not substituted for the common continuation. The target is the least implementation cost subject to every restart’s operating-value shortfall being at most a prescribed allowance. Implementation cost can charge departures from an installed rule and can include an administratively specified realized-action charge. A fixed organizational cost of administering lotteries is a separate contract term.

The central result is a theorem–algorithm connection. First, we identify exactly what the restart-price family optimizes. Its supremum is the dual of an occupation-flow relaxation in which an edge may allocate the successor regret budget differently from other edges arriving at the same state. The correction in the price Bellman recursion is therefore not a free-standing strong-duality theorem for the original constrained problem. Adding common successor-budget equations makes the occupation formulation exact. A rational two-period example has a constant-multiplier value of $5/4$, a restart-price supremum of $27/16$, and an original common-policy optimum of $7/4$. The equalities locate both the gain from restart prices and the coupling which the global algorithm must restore.

Second, we construct both sides of the interval from primitive inputs. Directed Bellman enclosures are computed before any price or policy is proposed. A backward conditional-cost minimization supplies an initial feasible policy; a local numerical proposal is repaired against its actual continuation and the upper operating enclosure. Price proposals use only the lower and upper operating enclosures. Two different bilinear formulations use one common set of state–date probabilities. An independent rational reader reconstructs the relevant inequalities and the entire covering tree. The stopping theorem accounts for the enclosure width, policy-box diameter, and deployment quantization. Exact operating ties create no inverse-gap singularity in this argument. The sufficient tree bound is exponential, and the experiment reports rather than conceals its consequences.

Third, a restriction–extension theorem gives a matched controlled-continuum result for a finitely observed initial population. With finitely many transition atoms per action, the set of states reached from that population under *all* action and shock sequences is finite at a finite horizon. Restriction to that set preserves the operating problem. More importantly, every feasible common policy on the set extends to a feasible Borel policy on the entire state space by choosing an operating-best action against the already extended continuation off the set. This proves equality of the finite and continuum revision optima, not merely validity of one endpoint. The dynamics may move the continuous state in an action-dependent and noninvertible manner. No autonomous type label or transition smoothing is required.

This result has an exact boundary. A finite observed initial law is the objective law, not an empirical approximation to an unobserved uniform population. Atomic laws need not approximate a continuous law in the topology required for this all-restart constrained optimization. Accordingly, the two observed-fleet calculations below do not replace the original uniform-initial-law cohort. The thirty positive-cost intervals in that cohort remain open. The broad Borel upper-density theorem and the structural aggregation, reset, and fiber results retain their separate hypotheses and are preserved in the technical volumes.

The new evaluation is generated only after the final five numerical source files and protocol are frozen on the remote repository. Twelve primary environments vary state count, horizon, action count, discounting, allowance, initial occupancy, and tie geometry. Each method constructs its own enclosures and incumbent. Neither receives a saved policy, an exact operating value, or an earlier price. At the absolute target 10^{-3} , both formulations meet four of twelve primary targets, including one zero-cost case; two positive-cost closures require branching. On all eight unclosed cases the price formulation returns a smaller certified interval, but it produces no additional primary target hit in this sample. The separate observed-fleet problems have horizons two and three and positive costs. Both are certified under the registered target, with the price formulation’s widths below 10^{-8} . These results show a complete working pipeline and an exact continuum transfer in the stated class; they do not establish uniformly practical global optimization.

1.1. *Relation to constrained-control duality and approximation*

Occupation measures express conventional expected constrained Markov decision problems (Altman, 1999). Finite linear-programming approximations can converge, with quantitative rates under suitable assumptions (Dufour and Prieto-Rumeau, 2013, Saldi, 2018). Our all-restart constraint is different from an initial-distribution average of regrets. Theorem 1 derives the precise occupation program corresponding to the proposed prices, rather than inferring an equivalence from the common use of multipliers. Its common-budget completion is a finite nonconvex formulation of the original policy object. Theorem 2 shows that dropping this completion changes the value even with action-independent transitions.

Information-relaxation duality relaxes information restrictions and uses penalties to recover bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). The relaxation derived here instead separates incoming allocations of a shared successor regret. It does not relax nonanticipativity, and the policy remains Markov throughout. Constant prices reproduce an initial-regret Lagrangian; state-date prices can be strictly stronger, but their finite supremum is still the identified relaxation. This is a formal relation to established dual objects, not a claim to a new general principle of duality.

The lower-node constructions use classical McCormick inequalities (McCormick, 1976); the direct-value baseline uses aggregate continuation products. The enhanced construction uses signed witness-relative regret and disaggregated products with simplex identities. Both are read by the same independent arithmetic checker and searched by the same driver. This isolates a formulation comparison, not an externally authored solver competition. Verified residual correction follows the separation between numerical proposals and mathematical certificates emphasized by Rump (2010). The previously favorable native SCIP comparisons remain in the supplement with their uncertified lower bounds distinctly identified.

The continuum theorem below is not a generic state-quantization theorem. It is an exact forward-closure result, together with a globally feasible continuation extension. It complements rather than subsumes the approximation results of Dufour and Prieto-Rumeau (2013) and Saldi (2018). The economic example is a designed observed-fleet service contract. Unlike the estimated replacement application of Rust (1987), it supplies neither estimated primitives nor empirical implementation costs. High-dimensional operating approximation of the kind studied by Brumm and Scheidegger (2017) remains an important separate test; the current enclosures use finite directed Bellman passes.

2. THE COMMON-POLICY REVISION PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$, and $t = 0, \dots, T - 1$. The known transition kernels P_t^a are Borel. Rewards, nonnegative implementation costs, and terminal rewards are Borel and satisfy

$$|r_t(x, a)| \leq R, \quad 0 \leq k_t(x, a) \leq K, \quad |g(x)| \leq G.$$

Throughout, $0 < \beta < 1$, $T < \infty$, and $\varepsilon > 0$. Define $S_h = \sum_{j=0}^{h-1} \beta^j$ and $S_0 = 0$. A policy p is a family of Borel probability vectors $p_t(\cdot | x)$. Its operating and implementation values and the optimal operating value satisfy

$$\begin{aligned} J_T^p &= V_T = g, & C_T^p &= 0, \\ V_t(x) &= \max_a \{r_t(x, a) + \beta P_t^a V_{t+1}(x)\}, \end{aligned} \tag{1}$$

$$J_t^p(x) = \sum_a p_t(a | x) \{r_t(x, a) + \beta P_t^a J_{t+1}^p(x)\}, \tag{2}$$

$$C_t^p(x) = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}.$$

For a specified initial probability law ν , the target is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \text{ for every } t, x\}. \tag{3}$$

An operating-optimal Borel policy exists by backward first-maximizer selection, so this feasible set is nonempty. We use an infimum when a general Borel optimizer has not been established. Finite models have a compact policy simplex and continuous constraints, hence attain the minimum.

Set

$$d_t(x, a) = V_t(x) - r_t(x, a) - \beta P_t^a V_{t+1}(x), \quad D_t^p = V_t - J_t^p. \quad (4)$$

Then $d_t \geq 0$, $D_T^p = 0$, and $D_t^p = \sum_a p_t(a | x)(d_t(x, a) + \beta P_t^a D_{t+1}^p(x))$. The economic restriction is $0 \leq D_t^p \leq \varepsilon$ everywhere. Its initial-law analogue, $\int D_0^p d\nu \leq \varepsilon$, is weaker and will be used only as a comparison.

For an installed rule a^0 , a representative cost is $k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}$. Physical switching from the previous action is a different friction and requires augmenting the state. A fixed lottery-administration fee f can be compared with a certificate in a designed budget exercise, but is not silently included in k . Scaling implementation costs by $c > 0$ scales K_R, L, U , and an appropriate absolute target by c , while leaving the operating constraint unchanged.

3. WHAT RESTART PRICES OPTIMIZE

For bounded nonnegative Borel prices λ_t with $\lambda_T = 0$, define $u_T = 0$ and

$$u_t(x) = \min_a \{k_t(x, a) + \lambda_t(x)(d_t(x, a) - \varepsilon) + \beta P_t^a[u_{t+1} + \varepsilon \min\{\lambda_t(x), \lambda_{t+1}(x)\}](x)\}. \quad (5)$$

Every feasible common policy satisfies $C_t^p + \lambda_t(D_t^p - \varepsilon) \geq u_t$. Indeed, backward substitution leaves the nonnegative hinge

$$(\lambda_t(x) - \lambda_{t+1}(y))_+ D_{t+1}^p(y) + (\lambda_{t+1}(y) - \lambda_t(x))_+ (\varepsilon - D_{t+1}^p(y)).$$

Together with $C_t^p \geq 0$, this proves the lower bound

$$L(\lambda) = \int \max\{0, u_0\} d\nu \leq K_R(\nu, \varepsilon). \quad (6)$$

The same induction permits lower disadvantages in place of d , an observation used by the implemented witness-only method.

3.1. An exact finite occupation interpretation

For this subsection let $X = \{1, \dots, n\}$. Write $P_{tia,j}$ for the transition probability and let μ be any nonnegative initial subprobability. For nonnegative variables y_{tia} define $z_{ti} = \sum_a y_{tia}$ and $f_{tij} = \beta \sum_a y_{tia} P_{tia,j}$. Edges only run from date t to date $t+1$, for $t < T-1$. Set missing edge sums to zero. Consider the linear program

$$\mathcal{O}(\mu) = \min_{y, \alpha, \gamma \geq 0} \sum_{t,i,a} k_{tia} y_{tia}, \quad (7)$$

$$z_{0i} = \mu_i, \quad z_{ti} = \sum_h f_{t-1,hi} \quad (t \geq 1),$$

$$\sum_a y_{tia}(\varepsilon - d_{tia}) \geq \sum_j \alpha_{tij} + \sum_h \gamma_{t-1,hi}, \quad (8)$$

$$\alpha_{tij} + \gamma_{tij} \geq \varepsilon f_{tij}. \quad (9)$$

The variables y are discounted occupations. The two nonnegative allocations on an incoming edge need not encode the same successor regret as allocations from other incoming states.

THEOREM 1—Price supremum and common-budget completion: *For every finite model satisfying Section 2,*

$$\sup_{\lambda \geq 0} \sum_i \mu_i u_{0i}(\lambda) = \mathcal{O}(\mu). \quad (10)$$

Both sides have finite attained optima. Adding variables $b_{ti} \in [0, \varepsilon]$, $1 \leq t < T$, and the equations

$$\alpha_{tij} = f_{tij} b_{t+1,j}, \quad \gamma_{tij} = f_{tij}(\varepsilon - b_{t+1,j}) \quad (11)$$

to (7)–(9) makes its optimum equal to the original all-restart common-policy value with objective measure μ . Thus price optimization solves an explicitly identified relaxation; restoring shared successor budgets solves the stated policy problem.

PROOF: Introduce free variables v_{ti} and nonnegative variables λ_{ti}, h_{tij} , with $v_T = \lambda_T = 0$. The price linear program maximizes $\sum_i \mu_i v_{0i}$ subject to

$$\begin{aligned} v_{ti} - \beta \sum_j P_{tia,j} v_{t+1,j} + (\varepsilon - d_{tia}) \lambda_{ti} - \beta \varepsilon \sum_j P_{tia,j} h_{tij} &\leq k_{tia}, \\ h_{tij} &\leq \lambda_{ti}, \quad h_{tij} \leq \lambda_{t+1,j}. \end{aligned} \quad (12)$$

There are no h variables in the last period. Given a price field, increasing h to the corresponding minimum can only relax these inequalities. Backward maximization of v then gives exactly (5). Conversely, (5) and those minima are feasible. This identifies the linear-program value with the supremum on the left of (10).

Assign dual multipliers y to (12) and α, γ to the two upper bounds on h . The free v variables give occupation conservation. The nonnegative λ variables give (8); the nonnegative h variables give (9). The resulting dual is precisely (7). It is feasible: for any feasible policy set $y_{tia} = \beta^t \Pr_\mu(X_t = i) p_t(a | i)$, $\alpha_{tij} = f_{tij} D_{t+1,j}^p$, and $\gamma_{tij} = f_{tij}(\varepsilon - D_{t+1,j}^p)$. At a positive-date node the budget inequality is an equality; at a root it is its all-restart regret inequality. The primal price program is also feasible, for example with zero prices and sufficiently small v . Its value is bounded by a feasible policy's cost. Finite-dimensional linear-program duality proves equality and attainment.

The same policy construction satisfies (11) with $b = D^p$, so the completed program has value at most the original optimum. In the reverse direction, take a completed feasible solution. At every node with $z_{ti} > 0$, put $p_t(a | i) = y_{tia}/z_{ti}$. Conservation ensures that a positive-probability transition of this policy goes to a positive-occupation node. For $t \geq 1$, substituting (11) in (8) and using incoming conservation gives

$$b_{ti} \geq \sum_a p_t(a | i) \{d_{tia} + \beta \sum_j P_{tia,j} b_{t+1,j}\},$$

with terminal budget zero. At roots of positive μ mass the same argument replaces b_{0i} by ε . Backward induction gives actual regret at most b at positive-date occupied nodes and at most ε at roots. At an unoccupied node choose a first maximizer of $r_t + \beta P_t J_{t+1}^p$ after the later policy has been completed. If the future policy is globally feasible, this yields $J_t^p \geq V_t - \beta \varepsilon \geq V_t - \varepsilon$. Occupied transitions do not enter unoccupied successors, so this extension does not change their values. It is globally feasible and has objective $\sum k y$. This proves the reverse inequality. *Q.E.D.*

The theorem does not say that a usual occupation-measure formulation is incapable of expressing common policies. It gives the additional bilinear compatibility which this particular price relaxation omits. The problem-specific content is the precise edge allocation and its completion, not linear-program duality itself.

COROLLARY 1—Clipping and constant multipliers: *In a finite model,*

$$\sup_{\lambda} L(\lambda) = \max_{S \subseteq \text{supp } \nu} \mathcal{O}(\nu \mathbf{1}_S). \quad (13)$$

For a constant price $\lambda_{ti} = \ell$ at all nonterminal dates, the un-clipped objective is

$$\sum_i \nu_i u_{0i} = \inf_p \left\{ \int C_0^p d\nu + \ell \left(\int D_0^p d\nu - \varepsilon \right) \right\}.$$

Consequently the best constant-multiplier bound is no larger than the un-clipped state–date–price supremum, which is no larger than K_R .

PROOF: For every vector v , $\sum_i \nu_i \max(0, v_i) = \max_S \sum_{i \in S} \nu_i v_i$. A finite maximum commutes with a supremum, so Theorem 1 proves (13). For the second statement, the price correction cancels successive constant-price allowance terms. Summing them along a trajectory leaves $-\ell\varepsilon$, while the disadvantages telescope to D_0^p . Backward minimization of the transformed objective proves the identity. Q.E.D.

The implementation uses a finite price cap and numerically proposed prices. It enumerates all nonempty masks when $n \leq 4$ and otherwise proposes the full mask and positive-mass singletons, subject to a declared time budget. Every proposed field is reconstructed rationally and ranked by its clipped certificate. A truncated mask search is not advertised as attaining (13); the earlier un-clipped-only objective is not silently equated with the displayed bound. Appendix A.2 gives an exact example in which the full-mask optimum is $3/8$ while a clipped field attains $15/32$.

3.2. A strict separation with exact rational values

THEOREM 2: *There is a two-state, two-period, two-action model with action-independent transitions for which the constant-multiplier optimum, the clipped restart-price supremum, and the common-policy optimum are respectively*

$$\frac{5}{4} < \frac{27}{16} < \frac{7}{4}. \quad (14)$$

The first two values have exact matching primal and dual witnesses. The final value follows from a one-variable affine minimization.

PROOF: Take $\beta = \varepsilon = 1/2$, zero terminal reward, and actions G, B with operating rewards $0, -1$. Every transition goes to state 1. At date 0 the cost of G is 1 in state 1 and 3 in state 2; at date 1 it is $3/2$ in either state. Action B costs zero. Let $\nu = (1/2, 1/2)$. Then $V = 0$ and $d(G) = 0, d(B) = 1$.

Let q denote the probability of B at date 1 in state 1 and q_i the two date-0 probabilities. Feasibility requires $0 \leq q \leq 1/2$ and $q_i \leq 1/2 - q/2$. Costs decrease with each q_i , so they equal these upper limits at an optimum. The objective becomes $7/4 + q/4$, minimized at $q = 0$ with value $7/4$.

Set $\lambda_0 = (1, 3)$ and $\lambda_1 = (3/2, 3/2)$. The recurrence gives $u_1 = (3/4, 3/4)$ and $u_0 = (9/8, 9/4)$, hence the lower value $27/16$. A feasible occupation witness in (7) takes the date-0 probabilities of B as $(1/4, 1/2)$ and the date-1 probability as $1/4$. The discounted occupations are $y_{01} = (3/8, 1/8)$, $y_{02} = (1/4, 1/4)$, and $y_{11} = (3/8, 1/8)$. On the two incoming edges use $(\alpha, \gamma) = (1/8, 0)$ and $(0, 1/8)$. Every occupation, node-budget, and edge inequality holds, and the objective is $27/16$. These matching witnesses prove the un-clipped supremum.

Clipping cannot improve it. With initial mass only at state 1 the original optimum is $9/8$ (choose $q = 1/2$); with mass only at state 2 it is $9/4$ (choose $q = 0$). Every price potential at the corresponding root is bounded by that positive optimum, hence so is its positive part. The same price field attains both bounds simultaneously.

Finally, for the expected-initial-regret problem choose $(q_1, q_2, q) = (0, 1, 0)$. Its average regret is $1/2$ and its cost is $5/4$. The constant price $\ell = 3$ gives $u_0 = (1/4, 9/4)$ and the same value. Weak duality proves the constant-multiplier optimum $5/4$. The conflicting preferred terminal randomizations explain the gap between the middle and final values: incoming edges cannot deploy different values of q at their common successor. *Q.E.D.*

The separation supplies a constructive reason to refine shared policy variables rather than assuming that more price proposals alone must close every interval. It also distinguishes a policy-to-certificate identity from a true optimization error. The former's clipping correction is explicitly negative: if its un-clipped nonnegative slack sum is $\mathcal{S}$, then $C^p(\nu) - L(\lambda) = \mathcal{S} - \int (-u_0)_+ d\nu$. We do not call this clipped expression a sum of nonnegative terms. The full earlier identity and its case-level diagnostics are retained in the supplement.

4. END-TO-END CERTIFICATION FROM OPERATING ENCLOSURES

The algorithm's inputs are the rational finite model, ε , an interval target $\eta > 0$, and a work budget. The only operating information supplied to price generation, candidate construction, or node relaxations is a pair of certified Bellman witnesses. Exact optimal operating values and inherited feasible policies are not inputs.

4.1. Enclosures and feasible policy generation

Let $\underline{V}, \bar{V}$ satisfy terminal bounds and the componentwise Bellman inequalities

$$\underline{V}_T \leq g \leq \bar{V}_T, \quad \underline{V}_t \leq \max_a (r_t^a + \beta P_t^a \underline{V}_{t+1}), \quad \bar{V}_t \geq \max_a (r_t^a + \beta P_t^a \bar{V}_{t+1}). \quad (15)$$

Backward induction gives $\underline{V} \leq V \leq \bar{V}$. Write $w_t = \|\bar{V}_t - \underline{V}_t\|_\infty$ and $\xi = \max_{t < T} w_t$. The implementation uses exact rational sums followed by outward dyadic rounding in each Bellman pass; it never computes an exact optimal V as an intermediate oracle. In representable tied examples, the two witnesses may coincide, which is disclosed rather than described as difficult approximation.

The two useful advantage quantities are

$$\underline{d}_{tia} = \max\{0, \underline{V}_{ti} - r_{tia} - \beta P_{tia} \bar{V}_{t+1}\} \leq d_{tia}, \quad (16)$$

$$d_{tia}^0 = \underline{V}_{ti} - r_{tia} - \beta P_{tia} \underline{V}_{t+1}. \quad (17)$$

The second need not be nonnegative. For $B_t^p = \underline{V}_t - J_t^p$, it gives the exact signed recursion $B_t^p = \sum_a p_{tia} (d_{tia}^0 + \beta P_{tia} B_{t+1}^p)$, with the appropriate actual terminal payoff. The lower-node program uses $\underline{V} - \varepsilon \leq J \leq \bar{V}$, not a fiction that $\underline{V}$ is an exact optimal value.

To create the initial incumbent, proceed backwards. Given the already constructed continuation, put

$$Q_a = r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x), \quad Z_a = k_t(x, a) + \beta P_t^a \hat{C}_{t+1}(x).$$

Minimize $\sum_a p_a Z_a$ over the probability simplex subject to $\sum_a p_a Q_a \geq \bar{V}_t(x) - \varepsilon$. An extreme minimizer has at most two positive actions and is found by enumerating feasible pure actions and pairs straddling the threshold. This is a conditional greedy construction, not a claim of globally optimal implementation. It supplies a feasible policy without any earlier run. A separately timed local nonlinear proposal can improve it; every resulting candidate is repaired and checked exactly.

LEMMA 1—Upper-witness repair: *Put $s = (1 - \beta)\varepsilon$ and assume $\xi < s$. If a common proposal q satisfies $J_t^q \geq V_t - \varepsilon - \delta$ everywhere, a backward repair produces $\hat{p}$ with $J_t^{\hat{p}} \geq \bar{V}_t - \varepsilon$ and $J_t^{\hat{p}} \geq J_t^q$. Its maximum row total-variation displacement, before deployment quantization, is at most*

$$\min \left\{ 1, \frac{\delta + \xi}{\delta + s} \right\}. \quad (18)$$

Rounding every action except an operating-best continuation action down to a mesh h and assigning the remaining probability to that best action preserves feasibility and adds at most $(m - 1)h$ to this bound.

PROOF: At a given step evaluate the original row of q against the repaired future. By induction its value v is no smaller than J_t^q . The best continuation action has value $M \geq V_t - \beta\varepsilon$. Its margin above $\bar{V}_t - \varepsilon$ is at least $s - \xi > 0$. If v is below the threshold, mix the original row with that best action using the minimal fraction. If the threshold deficit is z , then $z \leq \delta + \xi$ and the fraction is at most $z/(z + s - \xi) \leq (\delta + \xi)/(\delta + s)$. Otherwise no mixture is needed. Values weakly increase at each step. Downward rounding of nonbest probabilities moves at most $(m - 1)h$ probability mass to a weakly better action, so preserves both value inequalities. *Q.E.D.*

For policies whose rows differ by at most ρ in total variation, uniform perturbation constants sufficient here are

$$D_J = 2(RS_T + G)S_T, \quad D_C = K \sum_{t=0}^{T-1} \beta^t S_{T-t}. \quad (19)$$

Then $\|J^p - J^q\|_\infty \leq D_J \rho$ and $|C^p(\nu) - C^q(\nu)| \leq D_C \rho$. These conservative constants follow by bounding the range of each continuation action value and iterating the one-step perturbation inequality. They require no action separation.

4.2. Two node formulations and exact endpoints

A node is a box for all nTm probabilities, intersected rowwise with the simplex. At a nonempty box, rectangular Bellman minimization and maximization give exact bounds $J^{\min}, J^{\max}, C^{\min}, C^{\max}$ for every policy in the box. Each extremum is itself attained by a common Markov selector because the box is rectangular across state–date rows. A node can be eliminated if its simplex intersection is empty or if $J_{ti}^{\max} < \underline{V}_{-ti} - \varepsilon$ somewhere. These are rational conditions, not numerical infeasibility flags.

The direct-value formulation writes the original Bellman equations in J, C and uses McCormick envelopes for each aggregate product $p_{tia}(r_{tia} + \beta P_{tia} J_{t+1})$ and its implementation analogue. The price formulation writes the signed $B = \underline{V} - J$ equations and disaggregates products by successor. It imposes $\sum_a (p_{tia} B_{t+1,j}) = B_{t+1,j}$ and the analogous cost identity. It additionally uses the certificate planes

$$C_{ti} + \lambda_{ti}\{B_{ti} + w_{ti} - \varepsilon\} \geq u_{ti}, \quad C_{ti} \geq \max\{0, u_{ti}\}, \quad (20)$$

where $w_{ti} = \bar{V}_{ti} - \underline{V}_{ti}$ and u is reconstructed with (16). Since $D_{ti}^p \leq B_{ti}^p + w_{ti}$, the first plane follows from the price inequality for every truly feasible policy. The variable shift and product identities do not change the underlying common-policy object. Neither formulation dominates the other as a general polyhedral statement.

For a node relaxation $\min c'z$ subject to $Az = b$, $Ez \leq f$, and $l \leq z \leq u$, rounded proposals y and $\tau \geq 0$ give the exact rational lower endpoint

$$y'b - \tau'f + \sum_j \min\{r_j l_j, r_j u_j\}, \quad r = c - A'y + E'\tau. \quad (21)$$

No dual feasibility tolerance is needed: the residual is explicitly paid. The node lower bound is the maximum of this endpoint, zero, the rectangular cost bound, the reconstructed price bound, and its inherited parent bound. An exact common feasible policy supplies the upper endpoint. The global lower bound is the minimum over the complete leaf cover, clipped above by that feasible upper value. The checker reconstructs each row and split independently of the optimizer.

4.3. A matched finite stopping result

THEOREM 3—Witness-only finite completion: *Let a retained policy box have maximum row total-variation diameter at most ρ . Suppose (15) holds, $\xi < s$, and its rectangular operating test does not eliminate it. Evaluating and repairing any box member gives a cost at most*

$$\Omega(\rho, \xi, h) = D_C \left[\rho + \min \left\{ 1, \frac{D_J \rho + 2\xi}{D_J \rho + \xi + s} \right\} + (m-1)h \right] \quad (22)$$

above the box's rectangular cost lower bound. Consequently, choose witnesses and deployment mesh with

$$D_C \{\xi/s + (m-1)h\} \leq \eta/64, \quad \xi < s. \quad (23)$$

Without work caps, midpoint best-bound search that bisects a widest coordinate at least every fourth depth, and evaluates a box member, terminates with a valid interval of width at most η . The statement holds for both node formulations, even when every numerical LP proposal is unavailable.

PROOF: A retained box has $J^{\max} \geq \underline{V} - \varepsilon$ everywhere. Any member q differs from its rectangular operating-maximizing policy by at most ρ per row, hence

$$J^q \geq V - \varepsilon - (\xi + D_J \rho).$$

Apply Lemma 1 with $\delta = \xi + D_J \rho$. The rectangular cost minimizer is another policy in the same box, so its objective differs from $C^q(\nu)$ by at most $D_C \rho$. The repair and deployment change the cost by at most D_C times their maximum row displacement. This proves (22). Relaxation strengthening and inherited lower bounds can only improve the interval.

Under (23), the limit of (22) as $\rho \downarrow 0$ is at most $2D_C\xi/s + D_C(m-1)h \leq \eta/32$. Choose a positive coordinate width w_* such that (22), with $\rho = mw_*$, is below η . There are $d = nTm$ coordinates. Along a path, after at most $4d\lceil \log_2(1/w_*) \rceil$ levels every coordinate has width at most w_* : the scheduled widest-coordinate splits alone suffice. Thus every deeper retained node already has a repaired feasible policy within η of its lower bound. While the global interval exceeds η , best-bound search selects a leaf below the current upper value minus η , which cannot be such a sufficiently refined node. Only finitely many shallower nodes exist. Exact elimination cannot remove a feasible solution. The asserted stopping follows. *Q.E.D.*

The node bound implicit in this proof is exponential in the sufficient depth. It is an accuracy guarantee, not a polynomial complexity claim. What changes relative to an idealized exact-oracle theorem is that the implemented pipeline supplies the witnesses and the feasible upper policy appearing in the proof, and its arithmetic level depends on the requested target.

For lower-price conditioning, the implementation starts at 8 bits and adds 8 bits until (23) holds and

$$\Lambda \sum_{t=0}^{T-1} \beta^t (w_t + \beta w_{t+1}) \leq \eta/64, \quad \Lambda = 4096. \quad (24)$$

The cap is fixed before model generation. Since the price field is generated only afterwards, this is an *ex ante* conditioning rule, not a favorable exact-model price reused with noisy values. Directed finite Bellman widths tend to zero as precision increases, so the rule terminates. Exact rational residuals still certify any proposed multipliers, irrespective of their quality. There is no claim that increasing precision alone eliminates optimization slack.

5. A PAIRED CONTROLLED-CONTINUUM CONSTRUCTION

Consider the same Borel problem (3), with transition kernels

$$P_t^a(x, \cdot) = \sum_{s=1}^S \omega_{tas}(x) \delta_{F_{tas}(x)}(\cdot), \quad \omega_{tas} \geq 0, \quad \sum_s \omega_{tas} = 1, \quad (25)$$

where the maps and weights are Borel. They may be action dependent, noninvertible, and state dependent. Let the *objective* initial law have finite support H_0 , and recursively set

$$H_{t+1} = \bigcup_{x \in H_t, a, s: \omega_{tas}(x) > 0} \{F_{tas}(x)\}. \quad (26)$$

Equal successors are identified, not copied into history-specific nodes. This is essential: two paths that reach the same state and date must use the same future policy.

THEOREM 4—Finite observation and global extension: *Under Section 2 and (25), with a finitely supported objective initial law, the unrestricted Borel common-policy value equals the finite common-policy value on the forward sets (26):*

$$K_R^X(\nu, \varepsilon) = K_R^H(\nu, \varepsilon). \quad (27)$$

Every feasible finite policy has an explicit Borel extension satisfying the operating constraint at every point of X , not just at the forward sets. Therefore a paired finite certificate $[L, U]$ is a paired certificate for the full-state problem with exactly the same endpoints. The construction uses at most $|H_0|(mS)^t$ states at date t and incurs zero spatial approximation error.

PROOF: All possible successors of a point of H_t belong to H_{t+1} , so the operating Bellman value on these sets equals the restriction of V^X by backward induction. Every globally feasible policy restricts to a feasible finite policy and has the same objective. This proves $K_R^H \leq K_R^X$.

For the reverse direction take any finite feasible policy p^H . Put $\hat{J}_T = g$. After defining the later extension, set

$$\hat{p}_t(\cdot | x) = \begin{cases} p_t^H(\cdot | x), & x \in H_t, \\ \delta_{a_t(x)}, & x \notin H_t, \end{cases} \quad a_t(x) = \text{first maximizer of } r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x). \quad (28)$$

Finite sets are Borel. Finite-action first-maximizer selection and the Borel kernels make the extension Borel. Values on H_t are exactly the finite values because all successors stay on the forward sets. They satisfy $\hat{J}_t \geq V_t - \varepsilon$. Off H_t , assuming global feasibility at date $t + 1$, one has

$$\hat{J}_t(x) = \max_a \{r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x)\} \geq V_t(x) - \beta \varepsilon \geq V_t(x) - \varepsilon.$$

This completes the induction. Starting under ν , the policy never leaves the finite forward sets, so its implementation objective is unchanged. Taking infima proves the reverse inequality. Each state has at most mS successors, which gives the size bound. No state or probability is perturbed, establishing the zero spatial error. *Q.E.D.*

COROLLARY 2—Executable paired accuracy: *Suppose the finite forward model can be constructed with exact rational arithmetic, including decidable equality of successor states. Apply Theorem 3 to that model and use (28) off the forward sets. This gives a computable paired hierarchy and a stopping rule $U - L \leq \eta$ for (3). A running maximum of certified lower endpoints and running minimum of feasible upper endpoints give monotone sequences converging to the same value when the work caps are removed and targets tend to zero.*

The extension is also operational for the affine rational model below. At any queried state, evaluate its finitely branching continuation and test membership in the finite rational forward sets. This computes an action without an exact global operating oracle. In a piecewise-affine implementation, ordinary max-affine breakpoints coexist with exceptional preimages of the assigned finite policies. Both counts may grow exponentially with the horizon. The theorem guarantees a representation and a matched certificate, not cheap symbolic compilation. The publication contract checker independently reconstructs all forward states, merges, transition rows, rewards, and the terminal folding used in the experiment.

5.1. An observed-fleet service contract with original controlled atoms

The continuous condition is $x \in [0, 1]$. Operating rewards are $r(x, a) = x - c_a$, with

$$(c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2, \quad \beta = 19/20.$$

An installed action 0 costs nothing to retain; either departure costs $1 + x$. The two shock outcomes for the three actions are

$$\begin{aligned} F_0(x) &\in \{3x/4, 3x/4 + 1/20\}, \\ F_1(x) &\in \{17x/20 + 1/10, 17x/20 + 7/50\}, \\ F_2(x) &\in \{x/10 + 4/5, x/10 + 17/20\}, \end{aligned}$$

with probabilities $3/5, 2/5$. These are the original affine controlled atoms, not a finite-state replacement for different dynamics. Condition affects reward, revision expense, and every controlled successor. We fix $\varepsilon = 1/100$ and the observed initial fleet law $\nu = (3/4)\delta_{1/5} + (1/4)\delta_{4/5}$, at horizons 2 and 3.

A service organization can interpret the statewide allowance as a warranty on remaining operating value whenever its rule is restarted, including off-fleet states; the objective weights the vehicles in the observed fleet. This is a specified decision contract, not evidence that any particular organization uses it. The extension in Theorem 4 is what makes the warranty global despite the finite objective population. Lottery administration is assumed permitted by that contract and is charged separately if nonzero.

To illustrate decision units fixed before evaluation, assign \$500 to one normalized implementation unit and consider budget levels \$100, \$250, \$500, and \$1,000. The two certified optimal costs are approximately \$634.80 and \$1,434.35. Thus the two-period task has a feasible certified policy below \$1,000 but none below \$500; the three-period task cannot satisfy a \$1,000 implementation budget under the stated operating guarantee. These conclusions are robust to the reported optimization intervals. They are not estimates of market maintenance costs or empirical welfare. The \$500 scale and budget grid are explicit illustrative primitives, and a different scale changes the monetary conclusions mechanically.

For horizon 2 the active forward-set sizes are 2, 11, 63; for horizon 3 they are 2, 11, 63, 367. The last transition's expected g is folded into the last-period reward exactly, avoiding terminal padding. The policy problem has respectively 13 and 76 active decision nodes. Dummy nodes used for rectangular storage carry zero costs and have no incoming transitions from active states. Their equations are checked separately; they do not stand for omitted physical states. Table II reports the paired intervals.

5.2. What is not an initial-law approximation

Equation (27) keeps ν fixed. It does not authorize replacing a uniform initial law by finitely many atoms. In particular, the earlier thirty positive-cost uniform-initial-law intervals are unchanged, and the earlier nonlinear two-state widths remain roughly 25–76% at allowance $1/2$. Their original results and proofs are reproduced in the retained technical material. The one-sided Borel density theorem likewise remains a one-sided theorem outside the hypotheses of Theorem 4. A general matched numerical hierarchy for diffuse initial laws with controlled moving atoms requires further structure and is not inferred from finite observation.

6. FROZEN END-TO-END EVALUATION

6.1. Information, cases, and work accounting

The final numerical source and the model generator were frozen at commit `b636c40609863b498658db9d404676a5f74a236b`, before generating the twelve primary environments or the two fleet inputs. The development smoke test used a separate two-state model. The primary cases combine warranty maintenance, inventory, queueing, and time-varying ties, at three sizes:

$$(n, T, m) \in \{(3, 4, 3), (5, 8, 3), (8, 12, 4)\},$$

$$\beta \in \{7/8, 15/16, 63/64\}, \quad \varepsilon \in \{1/50, 1/100, 1/200\}.$$

The parameters vary jointly; these rows do not identify a separate causal effect of each dimension. Nonuniform initial probabilities are used, with the largest level placing mass only on two states.

TABLE I
ALL TWELVE SOURCE-FROZEN PRIMARY CASES

Case	Price upper	Direct width	Price width	Price rel.%	P nodes	P stop
W0	3.110700	0.000995	0.000987	0.0317	5	target
W1	8.236030	0.128	0.0126	0.1535	169	time_cap
W2	8.926491	0.0447	0.0111	0.1243	27	time_cap
I0	1.982232	0.000997	0.000892	0.0450	9	target
I1	5.044754	0.0509	0.0123	0.2440	203	time_cap
I2	10.244317	0.0718	0.0109	0.1061	23	time_cap
Q0	0.230546	9.3e-09	9.3e-09	0.0000	1	target
Q1	7.424602	0.0238	0.00237	0.0320	191	time_cap
Q2	10.652032	0.0506	0.0117	0.1102	29	time_cap
T0	1.074850	0.151	0.00855	0.7953	255	node_cap
T1	0.000000	0	0	0.0000	1	target
T2	6.135079	0.95	0.0524	0.8537	39	time_cap

W, I, Q, and T denote warranty, inventory, queue, and ties. Indices 0–2 have the sizes listed in the text. The primary criterion is absolute width $\leq 10^{-3}$; *target* denotes only that criterion. Relative width divides by the upper endpoint (zero is displayed for the exact zero-cost row). Direct and price upper endpoints, all costs, and all trajectories appear separately in the supplement. Rounded table entries do not replace the exact endpoints used to decide success.

The tie family has nonzero, time-varying operating values and different optimal faces across states and dates; it is one generator family, not three independent applications. Its hidden construction potential is not passed to either solver. The three other families have genuinely nonzero witness widths. No primary outcome was used to alter the frozen solver.

Each method starts from the same raw model and independently constructs its own witnesses and initial policy. A conditional greedy policy and a local proposal are charged to that run. Prices are generated from the witnesses, never from exact-run prices. The methods have the same 30-second soft construction allocation, 255-node cap, 3-second local allocation, and a 3-second allocation for each LP proposal. The price allocation is capped at 5 seconds *within* construction. Exact certificate operations finish before a budget check. Reported all-in time additionally includes encoding, compression, and independent replay, and is not capped at 30 seconds. Thus the comparison has equal declared construction rules and measured cumulative costs, not an assertion that every run consumes exactly the same total CPU time.

The environment is Python 3.13 with NumPy 2.3.5 and SciPy 1.17.0, one numerical thread, on GitHub-hosted Linux runners. Family jobs use separate machines; direct and price runs are sequential within a family process. Process high-water memory is therefore not a method-isolated peak. The supplement retains that limitation and reports all original timings. Floating optimization is only a proposal mechanism; both lower endpoints and deployed policies have exact rational checks.

6.2. Primary results: tighter intervals but unchanged target counts

Table I reports the complete primary sample. Both methods have four target hits: W0, I0, Q0, and T1. T1 is zero cost, Q0 closes at the root, and W0 and I0 require branching. The price formulation needs respectively 5 and 9 nodes on the branching successes, compared with 69 and 63 for the direct formulation. All four successes are new inputs generated after the freeze, but the price formulation adds *zero* method-relative target hits on this sample. This is different from counting cumulative historical closures.

TABLE II

PAIRED CERTIFICATES FOR THE ORIGINAL AFFINE ATOMS WITH AN OBSERVED INITIAL FLEET

Case	Price upper	Direct width	Price width	Price rel.%	P nodes	P stop
F0	1.269602	2.28e-09	2.28e-09	0.0000	1	target
F1	2.868705	0.000367	8.27e-09	0.0000	1	target

F0 and F1 have horizons 2 and 3. Their common initial law is $(3/4)\delta_{1/5} + (1/4)\delta_{4/5}$. Theorem 4, not a state-quantization approximation, makes these intervals valid for the full Borel-state contract. Both methods generate their own incumbents. These are two designed contracts, not independent empirical applications or closures of uniform-initial-law rows.

All eight cases left open by the direct formulation remain open with prices, but each has a smaller price interval. The largest price widths are about 0.01110, 0.01086, and 0.01174 in W2, I2, and Q2. T2 still has width 0.05238, although below the direct width 0.95040. These are informative improvements, not successful applications of the 10^{-3} stopping rule. T0 also reaches the node cap with a nonzero width. Increasing precision cannot remove these optimization gaps: the tied witnesses already have zero width in exact dyadic arithmetic.

Across the twelve primary cases, the direct method uses about 286.26 seconds of measured all-in work and the price method about 340.52 seconds. The largest individual all-in time is about 53.42 seconds. The more expensive price relaxation trades fewer processed nodes for tighter lower values; it is not declared a uniform speed winner. Full per-stage costs and proof sizes are available in Supplement Tables S2–S4, while every recorded lower/upper improvement appears in the machine-readable trajectory file. The search refinement and feasible upper generation are actually executed; these rows are not a price-only attachment to old incumbents.

6.3. Observed-fleet intervals and the full-state interpretation

The price widths are approximately 2.28×10^{-9} and 8.27×10^{-9} ; the direct three-period width is about 3.67×10^{-4} . All four intervals meet the predeclared target. Their independent arithmetic checks concern the finite forward model, while the independently written contract check reconstructs the exact original affine maps and terminal folding. The extension proof transfers those checks to the unrestricted Borel-policy target. It is an analytic theorem, not a machine-checked proof of global Borel measurability. The zero spatial-error assertion follows from exact forward closure and should not be attributed to the finite reader alone.

6.4. Precision and verification boundaries

The adaptive rule chooses 32, 40, and 48 bits across the three non-tie size levels and 32 bits for the fleet cases. The tied construction is exactly representable at the starting 8-bit witness level; deployment still uses at least 32 bits. Every table reports price magnitude, the actual witness guard ratio, and the price-amplification budget. Large prices and high discounting impose a real enclosure requirement; the reported finite passes do not demonstrate a cheap high-dimensional learned or sparse-grid oracle.

All 28 proof objects are independently reconstructed from raw primitives, witnesses, prices, complete covers, and policies. The publication stage repeats this reconstruction without re-running the optimizers and adds model-contract and corruption checks. This separates source identity, finite arithmetic validity, analytic validity, and performance generalization. The two formulations are separately reconstructed but share a search driver and proposal routine; they are not independent external replications. The historical SCIP native lower bounds remain outside the rational portfolio and are preserved as numerical comparisons, not relabeled certified results.

6.4.0.1. *Post-execution diagnostics.* A separate diagnostic program computes exact finite operating values only after all frozen prices, policies, and proof objects have been produced. It checks the policy-to-certificate identity and the fixed-field witness-loss inequality on the fourteen price runs. These exact values are not inputs to any price proposal, policy proposal, repair, branching decision, or reported endpoint. The supplement reports the clipping correction separately from the three nonnegative slacks, and separates price-certificate width from the final tree width.

7. IMPLICATIONS AND REMAINING NUMERICAL QUESTIONS

The results identify the common continuation as the organizing constraint. Restart prices have a precise occupation-flow meaning; the missing shared budgets can be written explicitly and restored by a convergent finite refinement. A finite solver can construct both endpoints using only certified operating enclosures. For finitely observed initial populations and controlled finite-atomic dynamics, exact forward closure and global extension turn those finite intervals into paired full-state certificates.

The successful fleet calculations and the new source-frozen sample answer different questions. The former exercise a matched continuous-state theorem on original action-dependent atoms with an observed initial law. The latter show that an end-to-end witness-only implementation works without privileged exact-model discoveries, while exposing remaining failures under finite budgets. Four of twelve primary targets are met by each formulation, and eight remain open. These results do not imply polynomial search complexity, superiority to modern uncertified global solvers, or an empirical welfare conclusion.

The diffuse-initial-law continuous problem remains consequential. Its old intervals are retained without revision of their endpoints; finite observations are not used as a substitute initial law in those calculations. Broader settings involving continuous actions, unbounded primitives, learned kernels, or multiple operating constraints also require their own approximation and feasibility arguments. The inherited lower-only multicriterion extension is not promoted into an unproved general repair theorem. The present contribution keeps the original common-policy target and supplies an explicit algorithmic chain where the stated hypotheses make that chain complete.

APPENDIX A: ADDITIONAL ERROR AND DUAL DETAILS

A.1. Operating-witness sensitivity of a price field

Suppose u is reconstructed with exact disadvantage d , and $\underline{u}$ with (16), using the same nonnegative field. The inequalities

$$0 \leq d_{tia} - \underline{d}_{tia} \leq w_t + \beta w_{t+1}$$

follow directly from the operating enclosure (including the clipping at zero). Backward minimum stability yields

$$0 \leq u_{ti} - \underline{u}_{ti} \leq \Gamma_t, \quad \Gamma_t = \sum_{s=t}^{T-1} \beta^{s-t} \|\lambda_s\|_{\infty} (w_s + \beta w_{s+1}).$$

The positive-part map is 1-Lipschitz, so the same Γ_0 bounds the change in the displayed clipped lower endpoint. Any directed approximation of the price expectation adds its certified one-step error to this recursion. In the executed model the expectation is an exact rational finite sum. This is a perturbation result for a *fixed* field; the algorithm's cap-based rule is imposed before generating the field, so it does not presuppose stability of an exact-model price optimizer.

A.2. Clipping can change the optimizing price

The mask maximization is not merely a different display of the full-mask optimum. Modify the two-period example in Theorem 2 as follows. Keep $\beta = \varepsilon = 1/2$ and deterministic transitions to state 1. At date 0, rewards of G, B are 0, -1 , while costs are $(0, 0)$ in state 1 and $(3, 0)$ in state 2. At date 1, rewards of G, B are 0, $-1/4$ and costs are $(1, 0)$ in both states. Let $\nu = (3/4, 1/4)$ and $g = 0$. The operating value is zero.

Let l be the price at the reached terminal decision state. Its potential is

$$u_1(l) = \min\{1 - l/2, -l/4\}.$$

For this fixed l , the best root potentials use prices 0 and 3 and equal $u_{01} = u_1(l)/2$ and $u_{02} = 3/2 + \min(3, l)/4 + u_1(l)/2$. The first claim follows from the zero-cost good action; the second follows by checking the positive slope before root price 3 and the negative slope thereafter. Thus the optimized full-mask objective is

$$3/8 + \min(3, l)/16 + u_1(l)/2,$$

whose maximum $3/8$ occurs at $l = 0$. Its clipped value there is also $3/8$. But $l = 3$ gives potentials $(-3/8, 15/8)$ and clipped value $15/32 > 3/8$. The latter is the actual common-policy optimum: take the terminal bad action with probability 1, use the root bad action with probability $3/8$ in state 2, and use the good action in state 1. Its operating shortfalls obey all restart bounds, its objective is $15/32$, and the displayed certificate matches it. Hence optimizing the un-clipped objective can select a strictly inferior clipped certificate. Enumerating the relevant initial masks fixes this mismatch without altering the validity of any field.

A.3. A quantitative form of repaired Borel upper density

The inherited upper-density result can be stated with a finite error modulus, although not with a universal computable partition-size rate. Fix a globally feasible policy p and define finite dominating measures

$$\mu_0 = \nu, \quad \mu_{t+1} = \sum_{a=1}^m \mu_t P_t^a.$$

Their total masses are m^t . For any Borel proposal q , let $e_t(x) = \text{TV}(q_t(\cdot | x), p_t(\cdot | x))$ and $E_t = \int e_t d\mu_t$. Put $M = 2(RS_T + G)$, $s = (1 - \beta)\varepsilon$, $c = 1 + M/s$, and

$$A_t = cE_t + \frac{Mc\beta}{s} \sum_{v=t+1}^{T-1} (\beta c)^{v-t-1} E_v,$$

$$\mathcal{B}(E) = K \sum_{t=0}^{T-1} \beta^t S_{T-t} A_t.$$

Then exact continuation-best repair of q gives a globally feasible $\hat{p}$ with

$$|C_0^{\hat{p}}(\nu) - C_0^p(\nu)| \leq \mathcal{B}(E). \quad (29)$$

To verify this, write $a_t = \text{TV}(\hat{p}_t, p_t)$ and $\Delta_t = |J_t^{\hat{p}} - J_t^p|$. The threshold deficit before repair is at most $Me_t + \beta P_t^p \Delta_{t+1}$ because p is feasible. The best repaired continuation has margin at

least s , so the repair mass is at most this deficit divided by s . Therefore

$$a_t \leq ce_t + (\beta/s)P_t^p \Delta_{t+1}, \quad \Delta_t \leq Mce_t + \beta cP_t^p \Delta_{t+1}.$$

Integrating and using $\mu_t P_t^p \leq \mu_{t+1}$ proves $\int a_t d\mu_t \leq A_t$. The implementation performance-difference expansion and the range bound KS_{T-t} then give (29).

Rational simple proposals can approximate p in these finitely many $L_1(\mu_t)$ norms; the displayed expression gives value convergence after repair. But the dominating masses and factors c can grow rapidly, arbitrary Borel integrals need not be effectively evaluable, and a repaired simple proposal need not itself remain simple. Thus (29) is not a two-sided computable scheme for a diffuse initial law. The finite-observation theorem in the main text obtains such a scheme by exact closure, not by suppressing these difficulties.

A.4. Exact finite products and common budgets

For $p \in [l, u]$, $v \in [a, b]$, each product $z = pv$ obeys the sign-safe inequalities

$$\begin{aligned} z &\geq ap + lv - la, & z &\geq bp + uv - ub, \\ z &\leq ap + uv - ua, & z &\leq bp + lv - lb. \end{aligned}$$

The displayed four McCormick inequalities, not any sign shortcut, are reconstructed by the reader. Signed $B = \underline{V} - J$ variables are allowed. For a probability simplex, $\sum_a p_a v = v$ supplies an exact product conservation identity. These constraints preserve one continuation variable per successor. The occupation completion (11) is an alternative exact formulation, while the implemented probability-box search enforces the same compatibility through the original Bellman equations. The occupancy LP itself is not falsely described as complete for K_R .

APPENDIX B: ORGANIZATION OF RETAINED MATERIAL

The numerical supplement contains all 28 new rows, all recorded search trajectories, oracle conditioning, work components, the complete new source/test contract, and a compact assumption matrix. It also reproduces the complete R47 main article and supplement, which retain the strict-gap quadratic analysis, earlier finite searches, exact aggregation, reset transfer, observable fibers, original uniform-initial-law certificates, and nonlinear examples. A separate historical volume preserves the earlier R47 historical material. No earlier source, result, proof object, or review is deleted or modified. The preservation manifest checks that statement against the immutable reviewed baseline; preservation itself is not evidence for a new theorem or numerical target hit.

REFERENCES

- Altman, E. (1999): *Constrained Markov Decision Processes*. Chapman and Hall/CRC. [2]
- Brown, D. B., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [3]
- Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [3]
- Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85(5), 1575–1612. [3]
- Dufour, F., and T. Prieto-Rumeau (2013): “Finite Linear Programming Approximations of Constrained Discounted Markov Decision Processes,” *SIAM Journal on Control and Optimization*, 51(2), 1298–1324. doi:10.1137/120867925. [2, 3]

- McCormick, G. P. (1976): “Computability of Global Solutions to Factorable Nonconvex Programs: Part I—Convex Underestimating Problems,” *Mathematical Programming*, 10, 147–175. doi:10.1007/BF01580665. [3]
- Rump, S. M. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 287–449. [3]
- Rust, J. (1987): “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 55, 999–1033. doi:10.2307/1911259. [3]
- Saldi, N. (2018): “Finite-State Approximations to Discounted and Average Cost Constrained Markov Decision Processes,” arXiv:1807.02994. [2, 3]